When Probe *and* Gallery are Low Quality: Decreasing Accuracy and Increasing Demographic Disparities in 1:N Identification

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Abstract

Given good-quality probe and gallery images, one-to-many (1:N) face identification is highly accurate, with near-zero rank-1 False Positive Identification Rates (FPIRs) for galleries containing 12M persons with $1-2$ enrolled images each. However, operational deployments related to public safety and national security often involve low-quality probes and increasingly leverage large-scale galleries that may include low-quality web-scraped images. Low-quality probes have been shown to cause FPIRs up to $\sim$5% for males and $\sim$10% for females, even for small galleries. But the impact of gallery quality remains underexplored.

In this work, we evaluate probe and gallery degradation using two state-of-the-art matchers (ArcFace, AdaFace) and four demographic cohorts defined by race (Black, White) and gender (female, male). For each group, we select 3.4k probes and two mated gallery instances each, then apply systematic degradation with six levels of blur (from $\sigma = 1.5$ to 4.0) or resolution (from 42×42 to 18×18), yielding 194 total probe-gallery quality combinations.

We find that average FPIRs increase up to 22.4% for ArcFace and 4.6% for AdaFace. For severely degraded probes, this increase is non-monotonic: the highest FPIRs occur when the gallery is least or most degraded. FPIRs also differ substantially across demographics, with differentials up to 22.1% for ArcFace and 5.9% for AdaFace. In general, males have the lowest FPIRs (up to 21.2% for ArcFace and 4.2% for AdaFace) and females have the highest FPIRs (up to 35.5% for ArcFace and 7.9% for AdaFace).

1. Introduction

In controlled experimental scenarios, state-of-the-art face recognition algorithms excel in one-to-many (1:N) identification. In a 2019 NIST evaluation, the two top-performing commercial systems achieved rank-1 False Positive Identification Rates (FPIRs) of 0.13% and 0.37% for a gallery containing 12M recently enrolled images corresponding to

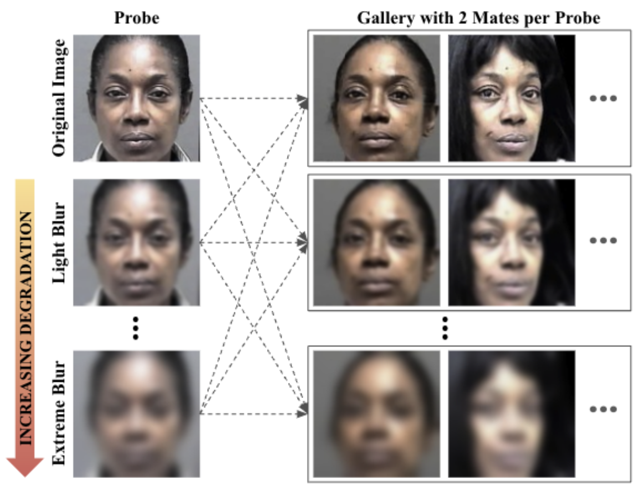

Figure 1. 1:N matching is performed across probe-gallery quality combinations, with systematically increasing degradation in each.

12M unique identities [16]. When an additional image was enrolled for each identity, FPIRs decreased slightly to 0.12% and 0.24%, respectively.

Importantly, FPIR decreases with the addition of images for existing identities, but *increases* with the addition of new identities to the gallery [8, 16, 33]. Thus, we would expect larger FPIRs for galleries containing more identities. Moreover, these near-zero FPIRs were obtained under conditions where both probe *and* gallery images were of high quality.

In operational deployments, however, such conditions are not guaranteed. Criminal investigations involving 1:N search, for example, often involve *low-quality* probe images [21], which challenges even state-of-the-art face recognition algorithms. Previous work on the use of degraded-quality probes reported FPIRs up to 10% for galleries containing $\sim$2.8k identities with one enrolled image each [8, 33]. Yet even these results rely on the assumption of a good-quality gallery, which may no longer hold in practice.

It is increasingly common for law enforcement agencies to leverage large-scale web-scraped galleries, with providers like ClearView AI providing databases containing upwards of 60 billion images sourced from unconstrained, publicly available web data [4, 9, 32, 37]. In 2024, ClearView AI reported approximately 2 million law enforcement searches of its database, nearly doubling the figure reported the previous year [11, 40].

In this emerging operational landscape, neither probe nor gallery quality can be assumed. Thus, the goal of this work is to characterize the individual *and* joint effects of probe and gallery image quality on 1:N identification performance.

2. Background and Related Work

Face recognition (FR) performance varies across multiple dimensions, with image quality representing a key source of variation [15, 24, 25]. Image quality components may be subject-related or capture-related. *Subject-related factors* are intrinsic to the human subject; facial occlusions, non-frontal pose, and extreme expressions reduce FR accuracy [7, 28, 35, 41, 43]. *Capture-related factors* arise from environmental and sensor conditions; poor illumination, blur, and low resolution likewise reduce accuracy [20, 34, 42, 44]. We focus on the two image-based quality factors of blur and resolution for the following reasons.

First, they are inherent to 1:N configurations relying on surveillance capture [22]. Operational probes are often extracted from still frames of surveillance video and are therefore likely to exhibit motion blur from subject movement, out-of-focus blur due to camera focus limitations, or both. Even for a high-resolution camera, an extracted probe typically represents a very small portion of the frame, and will likely have low resolution (see example in Fig. 2).

Second, blur and resolution are commonly simulated using high-quality imagery [17, 22], allowing other covariates to be controlled so that degradation effects can be examined in isolation. Real low-quality images from surveillance cameras are often more degraded than their simulated counterparts due to additional imaging artifacts and the difficulty of aligning low-quality faces for recognition [36]. Thus, the performance results obtained in this paper represent a lower-bound of the impact of blur and resolution.

Li et al. [22] provide a comprehensive overview of 1:N face recognition experiments addressing various resolution conditions for both probe and gallery images. However, their survey aggregates results from diverse studies with differing evaluation protocols, making direct comparison of recognition rates challenging. Additionally, the studies reviewed often consider fixed quality conditions rather than systematically varying image quality. Systematic variation of both probe and gallery quality is critical for operational relevance, as it allows identification of quality thresholds required for reliable recognition.

Our prior work [8, 33] was the first to systematically

Figure 2. Surveillance image from the Williams v. City of Detroit wrongful arrest case, along with a possible probe [12].

study image quality variation in 1:N face recognition and report FPIRs across race (Black, White) and gender (female, male). That work focused exclusively on probe degradation, simulating either increasing Gaussian blur ($\sigma = 1.0$–5.0) or decreasing spatial resolution (84×84 to 28×28).

Under mild degradations, FPIRs were near zero and roughly equivalent across all four demographic cohorts. Under severe degradations, FPIRs increased and clear gender disparities emerged. At $\sigma = 5.0$ blur, female FPIRs were approximately 1.3% for versus 0.4% for males. At 28×28 resolution, FPIRs were approximately 10% for females versus 7% for males.

Various algorithms have been developed specifically to handle very low-resolution inputs [10, 18, 23, 38]. Quality enhancement techniques (e.g., super-resolution) have also been proposed to improve degraded images, though these methods have shown mixed results in terms of identity preservation and suitability for FR. [30] Ultimately, the use of low-resolution-optimized algorithms or image enhancement methods in operational 1:N deployments remains unknown and is not standardized. For this reason, our experiments do not use specialized algorithms or pre-proccesing, to represent the "baseline" operational scenario.

2.1. Contribution of This Work

To our knowledge, no prior work has systematically evaluated the combined effects of probe *and* gallery degradation in a 1:N setting. Our study introduces a unified experimental framework that varies both probe and gallery across blur and resolution, using consistent evaluation protocols. This enables direct, within-study comparisons of matched (uniform) and mismatched (mixed) quality, reflecting realistic deployment scenarios wherein surveillance probes are matched against web-scraped galleries. We further extend prior work by analyzing demographic performance impacts under these dual-degradation conditions.

3. Overview of Experiments

3.1. Dataset and Degradations

We use the MORPH dataset [39], which contains mugshot images with frontal pose, neutral expression, consistent lighting, and plain background. The data is partitioned by race and gender—in this work, we use the Black female (BLF), Black male (BLM), White female (WHF), and White male (WHM) cohorts. To avoid demographic confounds noted in prior work [8, 33], we curate a balanced subset containing 3,400 identities per demographic. For each identity, we designate its most recent image as the probe and randomly select two remaining images as gallery instances, in keeping with the average number of gallery images per identity used by NIST [16]. This curation yields 3,400 probes and 6,800 gallery images per demographic.

The MORPH images vary in native resolution, ranging from 116×154 to 507×631, with a mode of 400×480. Each image is close-cropped about the face region, then aligned and resized to 112×112, as required for CNN ingestion. We then generate low-quality versions of each cropped image. For blur degradation, we use the NIST approach of applying Gaussian blur, with larger values of σ (the standard deviation parameter) corresponding to stronger blur [17]. For resolution degradation, we follow a standard downsample–upsample procedure used in prior work [8, 27, 33]: each image is downsampled then upsampled back to 112×112 using bicubic interpolation. (Since the cropped face regions are square, we simplify each resolution value as a single number, e.g., 112×112 $\rightarrow$ 112.)

To establish equivalent blur and resolution levels, we "self-match" (i.e., compute similarity scores between) each degraded image and its original, and pair blur / resolution levels that yield comparable similarity. Tab. 1 lists the six paired levels of blur and resolution. As self-match similarity is essentially a full-reference quality metric that would be infeasible for use in operational scenarios, we additionally include no-reference quality scores from MagFace [29], a top-performing metric in NIST's Face Analysis Technology Evaluation (FATE) Quality track [31].

Example degraded images are shown in Fig. 3 to images illustrate relative quality; as noted by NIST, perceived quality depends on display device and other factors [17].

3.2. Matching and Presentation of Results

We evaluate two state-of-the-art open-source face matchers: ArcFace and AdaFace. ArcFace [13], a combined-margin model trained on Glint360K(R100) [5] with weights from [2], has demonstrated performance comparable to commercial off-the-shelf systems used in real-world deployments. Because it is not designed for low-quality imagery, it represents operational systems that do not incorporate quality-specific enhancements. AdaFace [19], an adaptive-margin model trained on MS1MV2(R100) with weights from [1], incorporates quality-aware weighting and thus represents systems with some degree of quality-aware processing.

We perform 1:N matching experiments separately for blur and resolution. For each, the probe and gallery sets can each be in one of seven conditions: the original (undegraded) images or one of six degradation levels. This yields a 7 × 7 grid of probe–gallery combinations. Note that the baseline "original probe, original gallery" case is shared across both blur and resolution experiments.

Figure 4 presents heatmaps of average rank-1 FPIRs, aggregated across all demographics. We observe substantially higher error rates for ArcFace across all conditions—up to 18.2% compared to 4.6% for AdaFace. These contrasting results highlight that use of recent "quality-aware" CNN matchers is essential for operational scenarios involving lower-quality images. We also observe that, for each paired severity level, blur and resolution yield nearly identical patterns. For clarity and conciseness, subsequent figures therefore highlight AdaFace blur results as representative; tabular results include both models and both degradation types.

Table 1. Selected blur and resolution levels, with average "self-match" similarity scores (SM) and MagFace (MF) quality scores.

Applied Deg Level	Blur (σ)	Avg SM	Avg MF	Res (px)	IPD (px)	Avg SM	Avg MF
None	0.0	100	28	112	≈37	100	28
Light	1.5	92	27	42	≈14	92	27
Mild	2.0	86	26	35	≈12	87	26
Moderate	2.5	78	24	28	≈9	78	25
Strong	3.0	69	23	24	≈8	70	24
Very Strong	3.5	61	23	21	≈7	62	23
Extreme	4.0	51	23	18	≈6	52	23

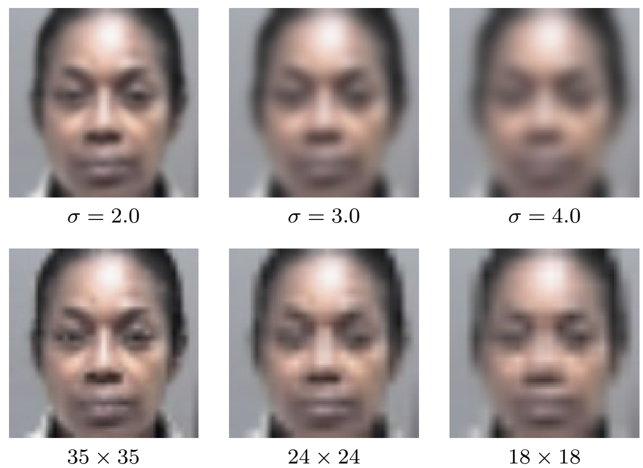

Figure 3. Ex. blur (top) and resolution (bottom) levels.

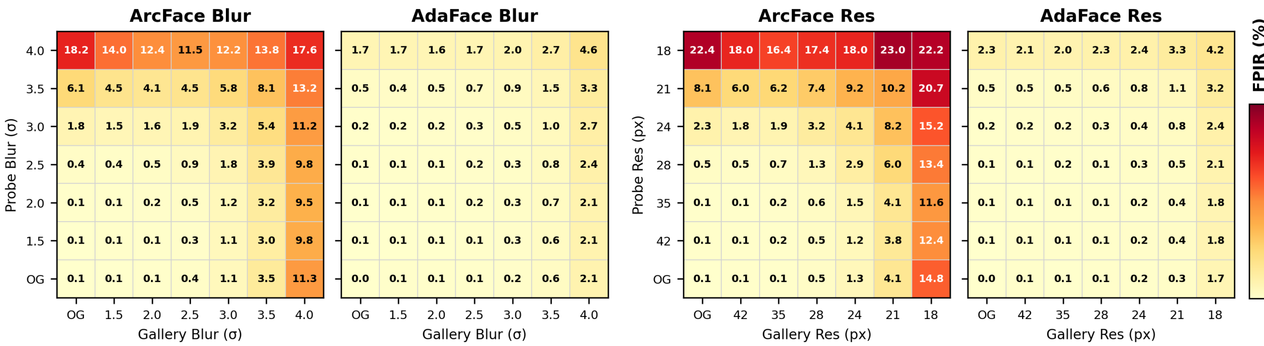

Figure 4. Average (demographic-aggregated) rank-1 FPIRs for ArcFace and AdaFace under blur or resolution degradations.

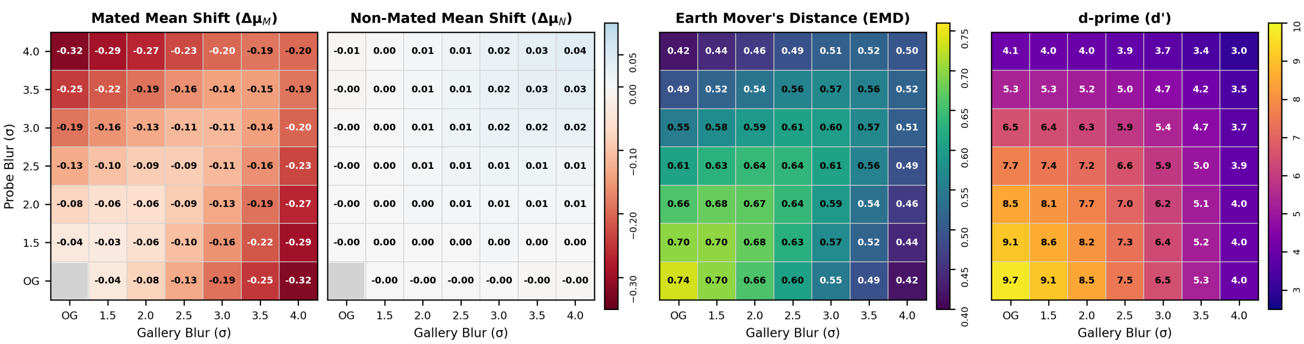

Figure 5. Average (demographic-aggregated) performance metrics for AdaFace under blur degradation.

4. Overall Performance Effects

The main goal of this work is to understand the impact of probe–gallery quality interactions on rank-1 FPIR, as shown in Fig. 4. To better understand the causes of FPIR variation, we examine the underlying score distributions.

To understand how the individual distributions shift, we calculate mated and non-mated mean shift ($\Delta\mu_M$, $\Delta\mu_N$) as the change in a distribution's mean similarity score versus the baseline. A negative $\Delta\mu_M$ value means mated scores shift to *lower* similarity, while a a positive $\Delta\mu_N$ value means non-mated scores shift to *higher* similarity—both indicate worse matcher performance.

To measure changes in separability of mated and non-mated distributions, we use two metrics. The first, d-prime (d'), measures the normalized distance between distribution means, accounting for pooled standard deviation. The d' metric is commonly used across FR studies. The second, Earth Mover's Distance (EMD, or Wasserstein distance), quantifies the minimum "work" required to transform one distribution into the other, accounting for the full shape of both distributions beyond just summary statistics. We include EMD as d' is technically only valid for Gaussian distributions. In conjunction, the two metrics help us understand the relative ease of distinguishing mated versus non-mated in a given condition—for both, lower values indicate worse separability and worse matcher performance.

Fig. 6 shows AdaFace score distributions for three representative blur levels (none / original, moderate blur, extreme blur) across the full range of gallery blur. Fig. 5 shows shift and separability heatmaps across the full blur matrix.

4.1. General Trends for Increasing Degradation

With lower quality in either probe *or* gallery, we observe consistent trends across both matchers:

Rank-1 FPIRs increase. Baseline FPIRs are near-zero for both matchers, and FPIRs for "low" and "mild" degradations remain low and comparable across matchers (0.1–0.2%). However, as degradation severity increases, matcher performance diverges substantially. ArcFace FPIRs rise to 18.2% for blur and 23.0% for resolution degradation, while AdaFace FPIRs reach only 4.6% and 4.2% respectively—roughly 4–5× lower than ArcFace at equivalent degradation levels.

Average mated scores shift substantially lower. The means of the mated distributions decrease significantly (up to ∼0.4 for ArcFace and ∼0.3 for AdaFace) at the most severe degradation levels. We also observe a slight "flattening" effect due to increased spread.

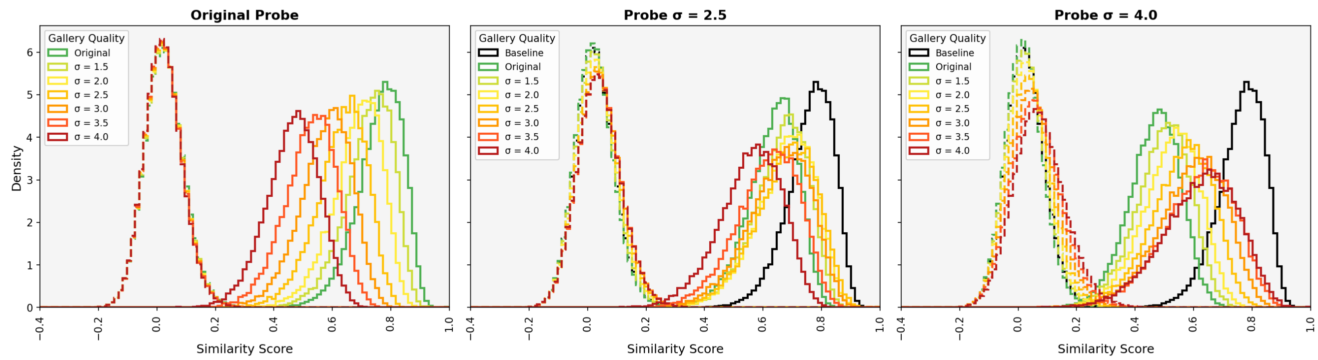

Figure 6. Ex. AdaFace score distributions for original probes or probes with "moderate" or "extreme" blur matched against a gallery with increasing blur. Note that "Baseline" indicates the original probe, original gallery condition.

Average non-mated scores shift slightly higher. The means of the non-mated distributions increase slightly (up to 0.11 for ArcFace and 0.04 for AdaFace) when both probe and gallery are degraded. We also observe a significant "flattening" effect due to increased spread.

Distribution separability decreases. Baseline EMD values of $\sim$0.7 decline to $\sim$0.3–0.4 at severe degradations, primarily due to downward-shifting in mated scores: mated means shift 3–4$\times$ more than non-mated means. The d' values, which normalize by pooled standard deviation, also indicate that score variance increases with degradation.

4.2. Thresholds for Probe-Only Degradation

In the traditional law enforcement 1:N deployment, degraded probes are matched against a good-quality gallery (e.g., a database of controlled-capture driver's license images). Here, we assume a good-quality gallery (i.e., "original") and analyze FPIRs across increasing probe degradation. Performance degradation was not gradual but exhibited cliff-like transitions at specific severity levels.

For ArcFace, FPIRs exceeded 1%, 5%, and 10% at the "strong", "very strong", and "extreme" degradation levels, respectively. The largest single-step increase occurred at the "very strong" to "extreme" transition. The step from $\sigma = 3.5$ to $\sigma = 4.0$ blur produced a 12.14 percentage point (pp) increase (6.10% to 18.23%), and the step from 21×21 to 18×18 produced a 14.31 pp increase (8.05% to 22.37%). These final transitions alone account for more FPIR degradation than all milder steps combined.

For AdaFace, FPIRs only exceeded 1% for "extreme" degradations, with the largest single-step increase at the final transition. For blur, the "very strong" to "extreme" step produced a 1.19 pp increase (0.55% to 1.74%), and for resolution, a 1.79 pp increase (0.47% to 2.26%). While these increases are substantially smaller than ArcFace's, they follow the same pattern: performance deteriorates disproportionately at extreme degradation.

4.3. Probe Versus Gallery Quality

Having established performance thresholds for probe-only degradation, we now examine how probe and gallery quality interact. We consider three operational scenarios: symmetric degradation (both images share the same quality level), probe-only degradation (P$\rightarrow$G: degraded probe versus original gallery), and gallery-only degradation (G$\rightarrow$P: original probe versus degraded gallery).

Across both matchers, symmetric degradation accounted for the maximum FPIR at 5 of 6 blur levels and 4–5 of 6 resolution levels. However, for severely degraded probes, a non-monotonic pattern emerges: FPIRs are high at both extremes of gallery quality.

For ArcFace with the most-blurred probes ($\sigma = 4.0$), FPIR reaches 18.23% against the original gallery (maximum asymmetry) and 17.60% against $\sigma = 4.0$ galleries (symmetric degradation)—but only 11.45% against galleries with intermediate blur ($\sigma = 2.5$). Similar patterns occur for the lowest-resolution probes: 22.4% FPIR against original galleries, 23.0% against 18×18 galleries, but only 16.4% against 35×35 galleries. AdaFace exhibits this pattern only for extreme degradation, and all maxima occur in the symmetric case.

These findings suggest two distinct failure modes for severely degraded probes. Maximum asymmetry likely produces mismatched feature representations between probe and gallery, while symmetric degradation leaves insufficient discriminative information in both images. Intermediate gallery degradation appears to strike a balance: producing sufficiently compatible representations while retaining enough identity information. AdaFace's quality-weighted training produces more degradation-robust representations across both failure modes.

In operational deployments, probes typically originate from surveillance imagery while galleries contain controlled-capture images—making the P$\rightarrow$G scenario most relevant. Table 2 presents the ratio P$\rightarrow$G / G$\rightarrow$P

Table 2. FPIR ratios for probe-only vs. gallery-only degradation, where values >1 indicate higher probe-only impact.

	Blur		Resolution	
Level	ArcFace	AdaFace	ArcFace	AdaFace
Light	0.92	0.51	1.00	0.45
Mild	0.89	0.78	1.13	1.08
Moderate	1.02	1.06	1.11	1.00
Strong	1.57	0.87	1.77	1.00
Very Strong	1.76	0.86	1.96	1.40
Extreme	1.61	0.83	1.51	1.35
Mean	1.29	0.82	1.41	1.05
Median	1.29	0.84	1.32	1.04

at each degradation level; values greater than 1.0 indicate probe quality has the larger impact.

For ArcFace, probe degradation consistently produced worse outcomes. The mean ratio was 1.29 for blur and 1.41 for resolution, with similar medians—indicating a consistent pattern rather than outlier-driven effects. At severe blur ($\sigma = 4.0$), probe-only degradation yielded 18.23% FPIR versus 11.34% for gallery-only; at 18×18 resolution, 22.37% versus 14.84%.

AdaFace exhibited more balanced behavior. For blur, the mean ratio was 0.82 (gallery degradation slightly worse); for resolution, 1.05 (near-equivalence). This likely reflects AdaFace's adaptive margin mechanism, which adjusts learning based on image quality during training.

5. Demographic Differentials in Performance

Figure 7 presents per-demographic FPIR heatmaps for AdaFace under blur degradation. We observe a consistent trend across both matchers and degradation types: Black females (BLF) have the highest FPIRs, Black males have (BLM) the lowest, and White females / males fall between the two extremes. This produces a substantial female-male disparity but complicates a simple race-based interpretation.

Gender gap (Δ_{G}) and race gap (Δ_{R}) are computed across all probe-gallery conditions as follows:

$$\Delta_{\mathrm{G}} = \frac{\mathrm{FPIR}_{\mathrm{BLF}} + \mathrm{FPIR}_{\mathrm{WHF}}}{2} - \frac{\mathrm{FPIR}_{\mathrm{BLM}} + \mathrm{FPIR}_{\mathrm{WHM}}}{2} \quad (1)$$

$$\Delta_{\mathrm{R}} = \frac{\mathrm{FPIR}_{\mathrm{BLF}} + \mathrm{FPIR}_{\mathrm{BLM}}}{2} - \frac{\mathrm{FPIR}_{\mathrm{WHF}} + \mathrm{FPIR}_{\mathrm{WHM}}}{2} \quad (2)$$

Note that the gender gap (Δ_G) cleanly separates the two highest-FPIR groups (BLF, WHF) from the two lowest (BLM, WHM), making it a straightforward measure of disparity. The race gap (Δ_R), however, averages the best-performing group (BLM) with the worst-performing group (BLF), partially masking the true extent of demographic variation. We report both for completeness, but the gender gap more directly captures the dominant axis of disparity in our results.

Table 3. Gender (Δ_G) and race (Δ_R) gaps in FPIR. Values > 0 indicate worse performance for female / Black subjects.

	ArcFace		AdaFace	
Condition	Δ_G	Δ_R	Δ_G	Δ_R
Baseline	0.12	−0.12	0.07	−0.07
Extreme Deg. ($\sigma = 4$ Blur, 18×18 Res)				
Probe Blur	11.20	8.03	1.79	1.00
Probe Res	14.50	7.62	2.79	1.29
Gallery Blur	9.60	4.92	0.60	−0.16
Gallery Res	12.56	4.03	1.32	0.30
Both Blur	8.09	7.65	2.52	1.17
Both Res	7.01	8.75	2.65	1.74
Averages (All Probe-Gallery Conditions)				
Blur Only	3.64	2.34	0.67	0.03
Res Only	4.87	2.63	0.91	0.20
Blur + Res	4.25	2.48	0.79	0.12

Table 3 reports these values for the baseline case, for "extreme" degradation ($\sigma = 4.0$ blur or 18×18 resolution) in probe only, gallery only, or both, and averaged across all blur and resolution probe-gallery combinations separately and collectively. In the baseline case, when probe and gallery images are both good-quality, gender and race gaps are minimal (near-zero) for both matchers. However, across nearly all scenarios with severe degradation in the probe and / or gallery, demographic differentials emerge and amplify with increasing degradation severity.

When probe and gallery have asymmetric degradation, the two matchers exhibit different behavior. For ArcFace, probe-only degradation produces the largest gender gaps—11.20 (blur) and 14.50 (resolution) percentage points—exceeding even the gaps observed under symmetric degradation of both images (8.09 and 7.01). This is notable because probe-only degradation with a high-quality gallery represents the traditional law enforcement use case. For AdaFace, by contrast, probe-only gender gaps (1.79, 2.79) are comparable to symmetric degradation gaps (2.52, 2.65), suggesting that quality-aware training may help equalize demographic impact across asymmetric conditions. Gallery-only degradation produces the smallest gaps for both matchers, consistent with our earlier finding that probe quality has a disproportionate impact on FPIR.

6. Key Takeaways

Degradation increases false positive identifications. As image quality decreases, rank-1 FPIRs increase substantially. For ArcFace, probe degradation was the primary driver of this increase, with the highest FPIRs occurring

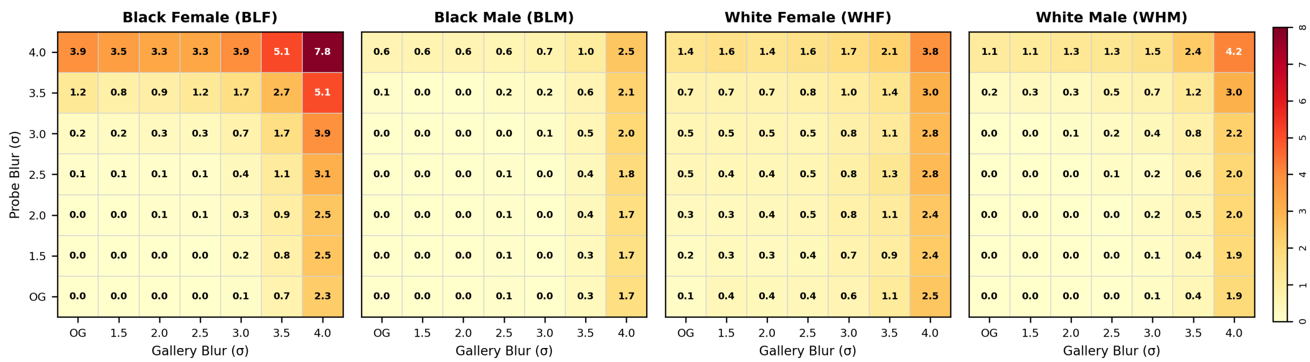

Figure 7. Per-demographic rank-1 FPIRs for AdaFace under blur degradation.

non-monotonically—for most-degraded probes matched against original-quality *or* equally-degraded galleries. For AdaFace, gallery degradation was more frequently the primary driver, with the highest FPIRs occurring uniformly for the equally-degraded probe-gallery case.

Quality-aware matchers are essential for low-quality probes. In law enforcement pipelines with good-quality galleries, probes are still likely to be low-quality surveillance images. For the most degraded probes, average rank-1 FPIRs reached ∼2% for AdaFace versus ∼22% for ArcFace—a 10× difference. However, even AdaFace's ∼2% corresponds to 272 mis-identified probes in our dataset. Operational scenarios with larger galleries and co-occurring degradations (pose, expression, illumination) will yield higher error rates; the FPIRs reported here represent a best-case scenario.

Demographic gaps in FPIR increase as degradation increases. Image quality degradation does not affect all demographic groups equally. Near-zero baseline gaps in FPIR suggest that demographic differentials are not inherent to the matchers' learned representations, but rather emerge from differential robustness to quality degradation. Gender gaps consistently exceed race gaps—often by 2–3×—though race gaps are attenuated by averaging the best-performing group (BLM) with the worst (BLF). Black females are disadvantaged on both dimensions, explaining their highest FPIRs; Black males are most robust—a pattern that defies simple race- or gender-based explanations and warrants further investigation.

Quality-aware training substantially reduces demographic disparities. Averaged across all probe-gallery conditions, ArcFace exhibits gender gaps of 3.6–4.9 percentage points and race gaps of 2.3–2.6 percentage points. AdaFace gaps are 5–6× smaller: 0.7–0.9 for gender and near-zero for race. This suggests that quality-adaptive training not only improves overall accuracy under degradation, but also mitigates demographic differentials.

7. Discussion and Conclusion

Near-zero 1:N reported for ideal 1:N conditions may lead practitioners in real-world deployments to expect similarly high accuracy. However, it is critical to understand that *operational* conditions may yield FPIRs that are much higher.

Probe images in law enforcement contexts are typically extracted from surveillance footage, where numerous quality degradations frequently co-occur. In our experiments, blur and resolution degradations were isolated and applied to images that were otherwise good-quality. For comparison, the probe from one documented wrongful arrest case (Fig. 2) had a MagFace score around 21—lower than even our "extreme" degradation levels, which averaged 22.8–22.9. Thus, the FPIRs reported here represent a lower bound on those that could occur operationally from real surveillance-quality probes.

Additionally, operational galleries may be substantially larger. Gallery size increases the difficulty of 1:N identification [6], with FPIR rising as enrolled identities increase [33]. NIST benchmark galleries contained 12–26.1 million images [16]; state law enforcement databases can exceed 50 million [3]. Larger still are web-scraped databases like ClearView AI, which reported over 60 billion images and 2 million law enforcement searches in 2024 [4]. Our findings show that low-quality galleries yield the highest FPIRs for quality-aware matchers and near-highest for non-quality-aware matchers, with demographic differentials reaching 7–9 percentage points.

Dual probe–gallery degradation represents an increasingly realistic operational scenario and induces a distinct failure mode not captured by probe-only studies. Fig. 8 illustrates score distributions for the most severe blur condition when degradation occurs in the probe only, the gallery only, or both. While probe-only or gallery-only degradation primarily affects the mated distribution—as commonly reported in prior work—dual degradation is the only scenario that produces a noticeable upward shift in non-mated scores, accompanied by substantial flattening of both mated and non-mated distributions. This behavior indicates increased uncertainty in similarity score assignment, rather than a simple loss of mated similarity. The effect is particularly pronounced for the non-quality-aware matcher (ArcFace), which exhibits both larger non-mated shifts and greater distribution overlap, while the quality-aware matcher (AdaFace) attenuates but does not eliminate this behavior. Importantly, the dual degradation case aligns closely with modern 1:N pipelines in which low-quality surveillance probes are matched against large, unconstrained, web-scraped galleries. Under these conditions, elevated FPIRs arise not only from collapsing mated scores but also from impostor similarity inflation, producing noisier candidate lists and weaker confidence signals for downstream human review.

Taken together, these factors create conditions where algorithmic FPIRs may be far higher than benchmark results suggest. Yet algorithmic error rates alone do not fully explain observed disparities in wrongful arrests. The majority of documented cases have involved Black males [26]—the demographic group with the *lowest* FPIRs in our experiments across nearly all conditions. This apparent paradox underscores the human element of 1:N identification pipelines: operator decisions, investigative practices, and systemic factors likely contribute to disparate outcomes beyond what algorithmic performance alone would predict.

In addition to human decision-making, gallery composition may further contribute to this apparent paradox. FPIR is known to increase as the number of enrolled identities increases: all else being equal, if a gallery contains a larger number of identities from one demographic group than another, probes from the overrepresented group have a higher likelihood of producing a false positive identification. Operational galleries are rarely demographically balanced: state driver's license databases broadly reflect state population demographics, while mugshot databases reflect historical arrest patterns within a jurisdiction. Such imbalances can therefore amplify false positive risk for specific demographic groups, independent of algorithmic accuracy–a phenomenon called the "watchlist imbalance effect" [14].

This work provides the first systematic evaluation of dual probe-gallery degradation in 1:N face identification, revealing non-monotonic error patterns that cannot be predicted from single-image degradation studies. Future work should examine additional degradation types and their interactions, evaluate whether quality-aware training can be optimized to minimize demographic differentials, and critically, investigate how image quality affects human operator accuracy and decision-making in the candidate review process.

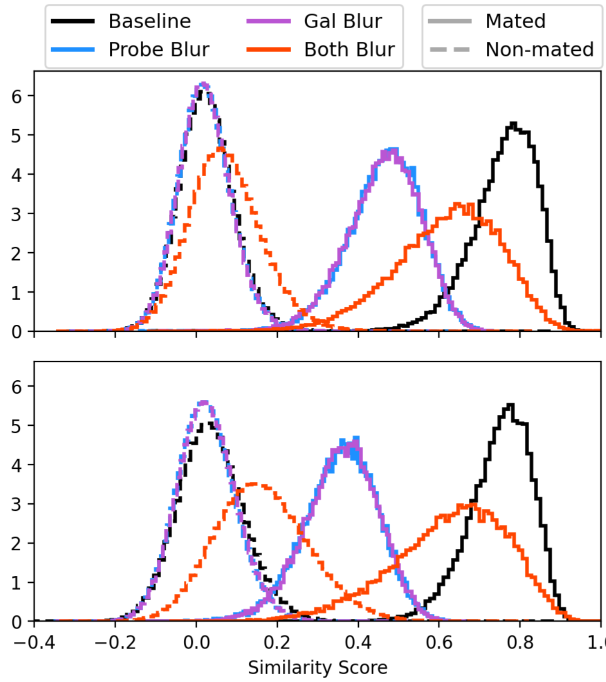

Figure 8. AdaFace (top) and ArcFace (bottom) score distributions for "extreme" blur ($\sigma = 4.0$) in probe, gallery, or both.

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